Stochastic Barrier Function Analysis of Market-Relative Wealth via Hybrid Switching Diffusions

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March 17, 2026

***Abstract*—We analyze the long-run behavior of a portfolio's log-relative wealth process $Z_t = \log(W_t/\mathcal{B}_t)$ under a state-dependent hybrid switching diffusion framework. The portfolio manager selects from three allocation modes — aggressive, moderate, and neutral — switching to recovery strategies with increasing probability as underperformance deepens below a threshold $-L$. Using the barrier function $V(z) = e^{\lambda(-z-L)^+} - 1$, which vanishes on the acceptable performance region $\{z \geq -L\}$ and grows exponentially below it, we prove that $\{Z_t\}$ is bounded from below in probability: $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \to 0$ as $R \to \infty$, with exponential decay rate. The proof combines the generalized Meyer–Ito formula, a Foster–Lyapunov generator condition, Gronwall's inequality, and Markov's inequality. Monte Carlo simulations with $n = 5{,}000$ paths over a horizon of $T = 20$ years confirm the theoretical bounds.**

***Index Terms*—stochastic differential equations, barrier functions, Lyapunov stability, hybrid switching diffusions, portfolio optimization, benchmark-relative performance, Ito calculus, Gronwall inequality**

I. Introduction

A central question in continuous-time portfolio theory is whether an active portfolio manager can ensure that relative performance — measured against a benchmark index — does not deteriorate without bound. While classical results address absolute wealth dynamics, the benchmark-relative setting introduces additional structure: the manager adapts strategy in direct response to underperformance.

This paper formalizes that adaptive response as a *hybrid switching diffusion*. The manager maintains a menu of three portfolio modes, switching among them based on the current log-relative wealth $Z_t = \log(W_t/\mathcal{B}_t)$. When the portfolio falls behind the benchmark by more than L, recovery modes are activated with state-dependent probability, generating upward drift to counteract the slide.

We analyze this system using the barrier function

$$V(z) = e^{\lambda(-z-L)^+} - 1, \quad \lambda > 0, \tag{1}$$

where $(-z-L)^+ = \max(0, -z-L)$. This function is identically zero for $z \geq -L$ and grows exponentially as $z \to -\infty$. It is a Lyapunov function in the classical sense: $V \geq 0$ everywhere and $V(z) = 0$ precisely on the target set $\{z \geq -L\}$, which is the equilibrium set of the controlled system.

Our main result (Theorem 6) establishes that under mild regularity conditions and a net-stability assumption on the switching policy, $\{Z_t\}_{t\geq 0}$ is bounded from below in probability, uniformly in time. The result is confirmed numerically in Section V.

II. Mathematical Preliminaries

A. Probability Space and Filtration

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$. The filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is an increasing family of σ-algebras ($\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$) encoding the information available at each time t.

B. Conditional Expectation and the Tower Property

Theorem 1 (Tower Property)**.** *For any integrable random variable X and sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}]\big]$.*

C. Brownian Motion and Ito Calculus

Definition 1 (Standard Brownian Motion)**.** *A process $\{B_t\}_{t\geq 0}$ is a standard Brownian motion if: $B_0 = 0$ a.s.; it has independent Gaussian increments $B_t - B_s \sim \mathcal{N}(0, t-s)$; and its paths are continuous a.s.*

Since $\mathbb{E}[B_t \mid \mathcal{F}_s] = B_s$, Brownian motion is a martingale. Its paths are nowhere differentiable, so the quadratic variation $\langle B \rangle_t = t$ is nonzero and the second-order Taylor term survives in any smooth functional expansion. Ito calculus replaces $(dx)^2 = 0$ with $(dB_t)^2 = dt$, $dB_t\, dt = 0$, $(dt)^2 = 0$.

Theorem 2 (Ito's Lemma)**.** *Let X_t satisfy $dX_t = a\, dt + \sigma\, dB_t$. For $f \in C^2$,*

$$df(X_t) = \Big(a f_x + \tfrac{1}{2}\sigma^2 f_{xx}\Big)dt + \sigma f_x\, dB_t.$$

The correction term $\frac{1}{2}\sigma^2 f_{xx}\, dt$ is absent in classical calculus and arises directly from $(dB_t)^2 = dt$.

D. Generalized Ito's Lemma for C^1 Functions

The barrier function (1) is C^∞ on each of the open regions $\{z > -L\}$ and $\{z < -L\}$, but only C^1 at $z = -L$ since the second derivative has a jump there. We therefore invoke the Meyer–Ito formula, which extends Ito's lemma to functions that are C^1 everywhere and C^2 away from a finite set of points.

Theorem 3 (Generalized Ito's Lemma, Meyer–Ito [1]). *Let $f : \mathbb{R} \to \mathbb{R}$ be C^1 everywhere and piecewise C^2 with a kink at c. Let L_t^c denote the local time of X_t at level c. Then*

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_0^t f''(X_s)\, d\langle X\rangle_s + \frac{1}{2}\Delta f'(c)\, L_t^c,$$

where $\Delta f'(c) = f'(c^+) - f'(c^-)$ is the jump in the first derivative.

E. The Lyapunov Generator

Applying Ito's lemma to a smooth $V(X_t)$ yields $dV = \mathcal{L}V\, dt + \sigma V'\, dB_t$, where the *Lyapunov generator* is $\mathcal{L}V(x) = aV'(x) + \frac{1}{2}\sigma^2 V''(x)$. Taking expectations (the stochastic integral has zero mean): $d\mathbb{E}[V(X_t)]/dt = \mathbb{E}[\mathcal{L}V(X_t)]$. If $\mathcal{L}V(x) < 0$, then V decreases on average — the process is attracted toward the zero set of V.

F. Gronwall's Inequality

Theorem 4 (Gronwall). *Let $m : [0,\infty) \to \mathbb{R}$ be continuous with constants $\alpha > 0$, $\beta \geq 0$ satisfying $m(t) \leq m(0) + \int_0^t (-\alpha\, m(s) + \beta)\, ds$ for all $t \geq 0$. Then*

$$m(t) \leq e^{-\alpha t} m(0) + \frac{\beta}{\alpha}(1 - e^{-\alpha t}), \quad \sup_{t\geq 0} m(t) \leq m(0) + \frac{\beta}{\alpha} < \infty.$$

G. Boundedness from Below in Probability

Definition 2. *A process $\{X_t\}_{t\geq 0}$ is* bounded from below in probability *if for every $\varepsilon > 0$ there exists $R > 0$ such that $\sup_{t\geq 0} \mathbb{P}(X_t \leq -R) \leq \varepsilon$.*

III. Market Model and Problem Formulation

A. Asset Dynamics

The market consists of a risky stock S_t, a benchmark index $\mathcal{B}_t$, and a risk-free asset with constant rate $r > 0$. Both assets follow geometric Brownian motion:

$$\frac{dS_t}{S_t} = \mu_s\, dt + \sigma_s\, dB_t^{(s)}, \qquad \frac{d\mathcal{B}_t}{\mathcal{B}_t} = \mu_m\, dt + \sigma_m\, dB_t^{(m)},$$

with $dB_t^{(s)} dB_t^{(m)} = \rho\, dt$ and constant parameters $\mu_s, \mu_m \in \mathbb{R}$, $\sigma_s, \sigma_m > 0$, $\rho \in [-1, 1]$. We assume continuous trading, self-financing portfolios, and no transaction costs.

B. Portfolio Wealth and Log-Relative Wealth

Let $w(t)$ be the fraction allocated to the risky asset (allowing $w > 1$ for leverage). Under self-financing, $dW_t/W_t = r\, dt + w(t)(\mu_s - r)\, dt + w(t)\sigma_s\, dB_t^{(s)}$. Define $Z_t := \log(W_t/\mathcal{B}_t)$. Applying Ito's lemma to each log and subtracting:

$$dZ_t = a(t)\, dt + \nu(t)\, d\widetilde{B}_t, \tag{2}$$

where $\widetilde{B}_t$ is a standard Brownian motion and

$$a(t) = r + w(t)(\mu_s - r) - \mu_m - \tfrac{1}{2}[w^2(t)\sigma_s^2 - \sigma_m^2], \tag{3}$$

$$\nu^2(t) = w^2(t)\sigma_s^2 + \sigma_m^2 - 2\rho w(t)\sigma_s\sigma_m. \tag{4}$$

When $Z_t > 0$ the portfolio outperforms; when $Z_t < 0$ it underperforms.

C. Three-Mode Switching Policy

The manager selects from three modes partitioned as $\mathcal{Q} = \mathcal{Q}_{\text{rec}} \cup \mathcal{Q}_{\text{neu}}$:

- *Aggressive* ($q = \text{agg}$, $w_{\text{agg}} = 1.5$): leveraged long, maximising upside drift in Z_t.
- *Moderate* ($q = \text{mod}$, $w_{\text{mod}} = 1.0$): fully invested, intermediate drift.
- *Neutral* ($q = \text{neu}$, $w_{\text{neu}} = 0.3$): defensive, low equity exposure.

The recovery set is $\mathcal{Q}_{\text{rec}} = \{\text{agg}, \text{mod}\}$ and the neutral set is $\mathcal{Q}_{\text{neu}} = \{\text{neu}\}$. For threshold $L > 0$, when $Z_t \leq -L$ a recovery mode is activated with probability at least $p(z)$, which increases as $z \to -\infty$. Under this policy the dynamics (2) become the *hybrid switching diffusion*:

$$dZ_t = a_q(Z_t)\, dt + \nu_q(Z_t)\, d\widetilde{B}_t, \quad q = q(Z_t) \in \mathcal{Q}. \tag{5}$$

a) Objective.: Prove that $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \to 0$ as $R \to \infty$.

IV. Barrier Function Analysis

A. Assumptions

Assumption 1 (Regularity). *For each $q \in \mathcal{Q}$, $a_q(z)$ and $\nu_q(z)$ are measurable, locally Lipschitz, and of at-most-linear growth.*

Assumption 2 (Adapted Switching). *The switching rule $q(\cdot)$ is measurable.*

Assumption 3 (Square Integrability). *For every $T < \infty$, $\mathbb{E}\Big[\int_0^T \nu_{q(Z_s)}^2(Z_s)\, ds\Big] < \infty$.*

Assumption 4 (Mode-by-Mode Stability). *There exist $\lambda, L, \eta > 0$ and $\kappa < \infty$ such that for all $z \leq -L$:*

- *Recovery modes $q \in \mathcal{Q}_{\text{rec}}$: $-\lambda a_q(z) + \frac{1}{2}\lambda^2\nu_q^2(z) \leq -\eta$.*
- *Neutral mode $q \in \mathcal{Q}_{\text{neu}}$: $-\lambda a_q(z) + \frac{1}{2}\lambda^2\nu_q^2(z) \leq \kappa$.*

Assumption 5 (State-Dependent Recovery). *There exists measurable $p : (-\infty, -L] \to (0, 1]$ with $\mathbb{P}(q(Z_t) \in \mathcal{Q}_{\text{rec}} \mid Z_t = z) \geq p(z)$ for all $z \leq -L$.*

Assumption 6 (Net Stability). *The effective recovery rate $\tilde{\eta}(z) := p(z)\eta - (1 - p(z))\kappa$ satisfies*

$$\tilde{\eta}^* := \inf_{z\leq -L} \tilde{\eta}(z) > 0 \iff \inf_{z\leq -L} p(z) > \frac{\kappa}{\eta + \kappa}.$$

B. The Barrier Function

We choose:

$$V(z) = e^{\lambda(-z-L)^+} - 1, \tag{6}$$

with λ as in Assumption 4. This satisfies: (i) $V(z) = 0$ for all $z \geq -L$ (zero on the target set); (ii) $V(z) > 0$ for $z < -L$ (positive in the downside region); (iii) $V(z) \to \infty$ as $z \to -\infty$ (exponential growth); (iv) $V \in C^1(\mathbb{R})$ (the Meyer–Ito formula applies).

C. Derivatives and the Ito Inequality

On $z > -L$: $V' = V'' = 0$. On $z < -L$: $V'(z) = -\lambda e^{\lambda(-z-L)}$ and $V''(z) = \lambda^2 e^{\lambda(-z-L)}$.

The one-sided derivatives at $z = -L$ are $V'((-L)^+) = 0$ and $V'((-L)^-) = -\lambda$, giving a jump $\Delta V'(-L) = 0 - (-\lambda) = \lambda > 0$. Applying Theorem 3 and dropping the non-negative local time term $\frac{\lambda}{2} L_t^{-L} \geq 0$ [1]:

Lemma 1 (Ito Inequality)**.** *With $\mathcal{L}_q V(z) = a_q V'(z) + \frac{1}{2}\nu_q^2 V''(z)$ and $M_t = \int_0^t \nu_q(Z_s) V'(Z_s)\, d\widetilde{B}_s$:*

$$V(Z_t) \leq V(Z_0) + \int_0^t \mathcal{L}_q V(Z_s)\, ds + M_t. \tag{7}$$

D. Generator Computation

Safe region ($z > -L$): $\mathcal{L}_q V(z) = 0$. The barrier contributes nothing above the threshold.

Downside region ($z < -L$): substituting the derivatives,

$$\begin{aligned}\mathcal{L}_q V(z) &= e^{\lambda(-z-L)}\Big[-\lambda a_q(z) + \tfrac{1}{2}\lambda^2 \nu_q^2(z)\Big] \\ &= (V(z)+1)\Big[-\lambda a_q(z) + \tfrac{1}{2}\lambda^2 \nu_q^2(z)\Big].\end{aligned} \tag{8}$$

The scalar bracket $[-\lambda a_q + \frac{1}{2}\lambda^2 \nu_q^2]$ is precisely what Assumption 4 controls per mode.

E. Foster–Lyapunov Drift Condition

Lemma 2 (Conditional Generator Bound)**.** *For all $z \leq -L$: $\mathbb{E}\big[\mathcal{L}_{q(Z_t)} V(Z_t) \mid Z_t = z\big] \leq -\tilde{\eta}^*(V(z)+1)$.*

Proof. At $z = -L$: $V(-L) = 0$ and $\mathcal{L}_q V(-L) = 0$, so the bound is trivial. For $z < -L$, (8) gives $\mathbb{E}[\mathcal{L}_q V \mid Z_t = z] = (V(z)+1) \cdot \mathbb{E}[\text{scalar} \mid Z_t = z]$. By Assumptions 4 and 5: with probability $\geq p(z)$ the scalar is $\leq -\eta$, and with probability $\leq 1 - p(z)$ it is $\leq \kappa$, so $\mathbb{E}[\text{scalar} \mid Z_t = z] \leq -\tilde{\eta}(z) \leq -\tilde{\eta}^*$. Since $V(z) + 1 > 0$ the result follows. □

Lemma 3 (Global Foster–Lyapunov Condition)**.** *There exists $d < \infty$ such that for all $z \in \mathbb{R}$:*

$$\mathbb{E}\big[\mathcal{L}_{q(Z_t)} V(Z_t) \mid Z_t = z\big] \leq -\tilde{\eta}^* V(z) + d. \tag{9}$$

Proof. For $z > -L$: $\mathcal{L}_q V(z) = V(z) = 0$, so (9) holds for any $d \geq 0$. For $z \leq -L$: Lemma 2 gives $\mathbb{E}[\mathcal{L}_q V \mid Z_t = z] \leq -\tilde{\eta}^*(V(z)+1) = -\tilde{\eta}^* V(z) - \tilde{\eta}^* \leq -\tilde{\eta}^* V(z)$, so (9) holds with $d = \tilde{\eta}^*$. □

F. Main Results

Taking the outer expectation of (9) via the Tower Property: $\mathbb{E}[\mathcal{L}_{q(Z_t)} V(Z_t)] \leq -\tilde{\eta}^* \mathbb{E}[V(Z_t)] + d$. Taking expectations in (7) ($\mathbb{E}[M_t] = 0$ by Assumption 3, Fubini for the integral):

$$\mathbb{E}[V(Z_t)] \leq V(Z_0) + \int_0^t \big(-\tilde{\eta}^* \mathbb{E}[V(Z_s)] + d\big) ds. \tag{10}$$

Theorem 5 (Uniform Barrier Moment Bound)**.** *Under Assumptions 1–6, $\sup_{t \geq 0} \mathbb{E}\big[e^{\lambda(-Z_t - L)^+} - 1\big] \leq C := V(Z_0) + d/\tilde{\eta}^* < \infty$.*

Proof. Apply Theorem 4 to (10) with $m(t) = \mathbb{E}[V(Z_t)]$, $\alpha = \tilde{\eta}^*$, $\beta = d$. □

Theorem 6 (Boundedness from Below in Probability)**.** *Under Assumptions 1–6, for all $R > L$,*

$$\sup_{t \geq 0} \mathbb{P}(Z_t \leq -R) \leq \frac{C}{e^{\lambda(R-L)} - 1} \longrightarrow 0 \quad \textit{as } R \to \infty.$$

Proof. On $\{Z_t \leq -R\}$ we have $V(Z_t) \geq e^{\lambda(R-L)} - 1 > 0$. Markov's inequality and Theorem 5 yield $\mathbb{P}(Z_t \leq -R) \leq \mathbb{E}[V(Z_t)]/(e^{\lambda(R-L)} - 1) \leq C/(e^{\lambda(R-L)} - 1)$. □

Corollary 1. *For large R, $C/(e^{\lambda(R-L)} - 1) \sim Ce^{-\lambda(R-L)}$, so the tail probability decays exponentially at rate λ.*

V. Numerical Simulation

We verify the theoretical results via Euler–Maruyama Monte Carlo simulation.

A. Parameter Specification

Table I summarises all parameters. The market parameters are standard continuous-time finance defaults. The neutral mode has negative drift $a_{\text{neu}} = -0.0196$ because $w_{\text{neu}} = 0.3$ is insufficient to overcome the benchmark's excess return; this makes Assumption 4 non-trivial, and the net stability condition (Assumption 6) is the binding constraint.

TABLE I
Simulation Parameters

Parameter	Symbol	Value
Stock drift / vol	μ_s, σ_s	0.10, 0.20
Benchmark drift / vol	μ_m, σ_m	0.08, 0.15
Correlation / risk-free	ρ, r	0.70, 0.03
Aggressive allocation	w_{agg}	1.5 (leveraged)
Moderate allocation	w_{mod}	1.0 (fully invested)
Neutral allocation	w_{neu}	0.3 (defensive)
Threshold / barrier exp.	L, λ	1.0, 0.075
Recovery prob. base	$p_{\min}$	0.70
Computed a_q, ν_q	$a_{\text{agg}}, a_{\text{mod}}, a_{\text{neu}}$	0.0213, 0.0113, −0.0196
	$\nu_{\text{agg}}, \nu_{\text{mod}}, \nu_{\text{neu}}$	0.2225, 0.1432, 0.1162
η, κ, $\tilde{\eta}^*$	—	7.86×10^{-4}, 1.50×10^{-3}, 9.9×10^{-5}
Bound	C	1.000
Paths / steps / horizon	n, N, T	5000, 20000, 20 yr

The recovery probability is $p(z) = \min(1,\ 0.70 + 0.30\,(-z - L)^+)$, reaching $p_{\min} = 0.70$ at the boundary $z = -L$ and ramping toward 1 as the portfolio falls deeper. One verifies $p_{\min} = 0.70 > \kappa/(\eta + \kappa) \approx 0.657$, so Assumption 6 holds with $\tilde{\eta}^* = 9.9 \times 10^{-5} > 0$. Within the recovery region, the probability of activating the aggressive mode ramps from 0.50 toward 0.70, while the moderate mode fills the remainder.

B. Sample Paths

Figure 1 shows eight representative paths over $T = 20$ years. All paths fluctuate freely above the threshold $-L = -1.0$ and periodically dip into the recovery region (shaded red). Crucially, no path drifts arbitrarily downward: each excursion below the threshold activates recovery modes with

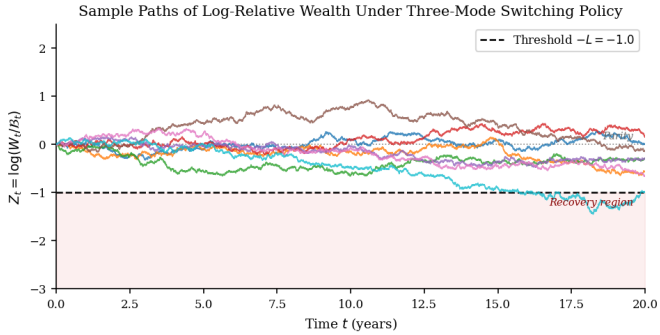

Fig. 1. Eight sample paths of $Z_t = \log(W_t/\mathcal{B}_t)$ under the three-mode switching policy. The dashed line marks the threshold $-L = -1.0$; the red-shaded region is where the policy activates recovery modes. No path drifts to $-\infty$.

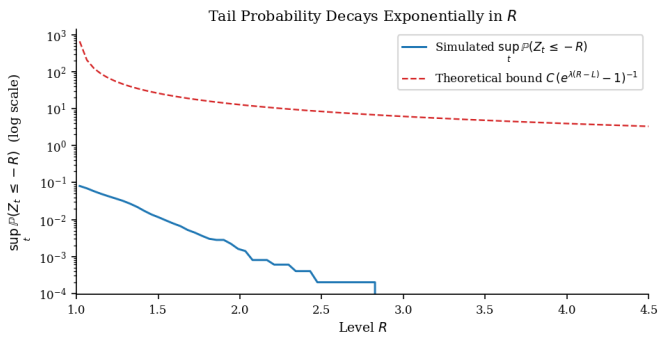

Fig. 2. Simulated supremum tail probability (solid) and theoretical bound (dashed) on a log scale. Both decay exponentially at rate $\lambda = 0.075$; the bound is valid at every $R > L = 1.0$.

increasing probability, generating the visible upward corrections. Across all 5,000 paths, the process spends approximately 2.4% of total time in the recovery region.

C. Tail Probability Decay

Figure 2 plots $\sup_t \mathbb{P}(Z_t \le -R)$ against the theoretical bound $C(e^{\lambda(R-L)}-1)^{-1}$ on a logarithmic scale. The simulated tail probability lies strictly below the theoretical envelope at every $R > L$, confirming Theorem 6. Both curves decay at the same exponential rate $\lambda = 0.075$ for large R, visible as equal slopes on the log-scale plot. Table II tabulates selected values; the large gap between simulation and bound quantifies the conservatism of the Foster–Lyapunov estimate.

TABLE II
SIMULATED VS. THEORETICAL TAIL PROBABILITIES

R	Simulated $\sup_t \mathbb{P}(Z_t \le -R)$	Bound $C/(e^{\lambda(R-L)}-1)$
1.5	0.01140	26.170
2.0	0.00140	12.840
2.5	0.00020	8.398
3.0	< 0.0001	6.179

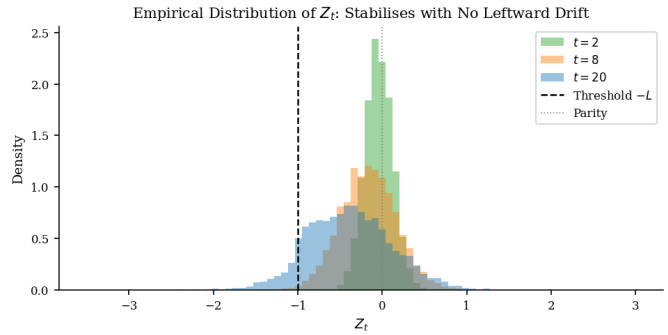

Fig. 3. Empirical distribution of Z_t at $t = 2, 8, 20$ years. The distribution stabilises rapidly; the left tail remains bounded with no drift toward $-\infty$, consistent with boundedness from below in probability.

D. Stationarity of the Distribution

Figure 3 shows the empirical distribution of Z_t at $t = 2, 8, 20$ years. After a short initial transient, the distribution stabilises and the histograms at $t = 8$ and $t = 20$ are nearly identical, suggesting convergence to a stationary distribution. No probability mass accumulates near $-\infty$ across any snapshot, which is precisely the content of Theorem 6. The distribution is positively skewed: the switching policy pushes Z_t upward from the downside region, but no corresponding mechanism caps the upside, allowing the right tail to spread freely over time.

VI. Financial Interpretation

Theorem 6 guarantees that the log-relative wealth $Z_t = \log(W_t/\mathcal{B}_t)$ cannot drift arbitrarily far below the benchmark with significant probability, regardless of the time horizon. For any tolerance $\varepsilon > 0$ there exists a floor $-R^*$ such that $\sup_{t\ge 0} \mathbb{P}(Z_t \le -R^*) \le \varepsilon$.

The barrier $V(z) = e^{\lambda(-z-L)^+} - 1$ formalises the manager's risk objective: it assigns zero cost to any $z \ge -L$ and exponentially growing cost to deeper underperformance. The generator analysis makes the mechanics precise: above the threshold $\mathcal{L}_q V = 0$, so no corrective action is needed, and the barrier is completely insensitive to normal market fluctuations. Only when $Z_t < -L$ does the negative drift condition engage, and the three-mode switching policy ensures that recovery dominates on average.

The three-mode structure captures a realistic graduated response: as underperformance deepens from moderate (z near $-L$) to severe ($z \ll -L$), the probability of activating the aggressive leveraged mode increases, providing stronger corrective drift. The net stability condition (Assumption 6) quantifies the minimum aggressiveness required: $p_{\min} > \kappa/(\eta + \kappa) \approx 0.657$. This is a hard lower bound on how often the manager must use recovery modes when below the threshold — a practically interpretable constraint that can be incorporated directly into a portfolio mandate.

VII. Conclusion

We have established that the log-relative wealth of a portfolio under a three-mode state-dependent switching policy is bounded from below in probability, with tail bound $\sup_{t\geq 0} \mathbb{P}(Z_t \leq -R) \leq C(e^{\lambda(R-L)} - 1)^{-1}$. The proof proceeds in five steps: (1) choose the barrier $V(z) = e^{\lambda(-z-L)^+} - 1$, vanishing on the target set $\{z \geq -L\}$; (2) apply the Meyer–Ito formula and drop the non-negative local time term; (3) establish the global Foster–Lyapunov condition using mode-by-mode stability (A4), state-dependent recovery probability (A5), and net stability (A6); (4) apply Gronwall's inequality to obtain $\sup_{t\geq 0} \mathbb{E}[V(Z_t)] \leq C$; (5) apply Markov's inequality for the exponential tail bound. Monte Carlo simulation with $5{,}000$ paths over 20 years confirms all theoretical predictions: the barrier expectation remains uniformly bounded well below C, the tail probability decays exponentially in R, and the distribution stabilises with no leftward drift.

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